TECHNICAL SPECIFICATION FOR 48V FLOAT CUM BOOST CHARGERS

1.0 SCOPE:

1.1. This specification covers the design, manufacture, testing, inspection and testing before supply and delivery at destination stores basis of following 48V Float Cum Boost Chargers with D.C. Distribution required for communication equipment at 400 KV /220KV / 132KV substations.

48V/35A, 50A & 100A Float Cum Boost Charger (1+0) with D.C. Distribution.

1.2. It is not intent to specify completely herein all the details of the design and construction of material. However, the material shall conform in all respects to high standards of engineering design and workmanship and shall be capable of performing in continuous commercial operation up to the Bidder's guarantee, in a manner acceptable to the purchaser, who will interpret the meanings of drawings and specification and shall have the power to reject any work or material which, in his judgment is not in accordance therewith. The offered material shall be complete will all components necessary for their effective and trouble-free operation. Such, components shall be deemed to be within the scope of Bidder's supply irrespective of whether those are specifically brought out in this specification and/or the commercial order or not.

2.0 STANDARDS:

2.1 The design, manufacture and testing of various equipments covered by this specification shall comply with the following Indian standards unless other-wise specified in the specification.

Indian standards	Title
Battery charger	
IS 7204(Part-1)/1974/1991	Stabilized power supplies DC output Part: Terms and definitions
IS 7204(Part-2)/1990/1991	Stabilized power supplies DC output Part-2: Rating and performance
IS 7204(Part-3)/1980/1991	Stabilized power supplies DC output Part-3 Radio frequency interference test
IS 7204(part-4)/1980/1991	Stabilized power supplies DC output Part-4 Tests other than for Radio Frequency interference
IS 3895/1961	Monocrytalline semiconductor rectifier cells and stacks
IS4540/1968/1991	Monocrytalline semiconductor rectifier Assemblies and equipment
DC Distribution	
IS5039/1983/1991	Distribution feeders for voltages not exceeding 1000V AC and 1200 V DC (first revision) (amendment I)

IS2675/1983/1991 (Second revision)	Enclosed distribution fuse board and output for voltages not exceed 1000V AC or 1200V DC
<u>Fuses:</u> IS9224(part-4/1980/1991)	Low voltage fuses part 4: Supplimentary requirement for fuse link for the production of semi conductor devices
IS13703(part-1)/1993 IEC 269-1/1986	Low voltage fuses for voltages not Exceeding 1000V AC or 1500V DC. Part:1: General requirement (superseding IS9224 (Part-1)/1979)

Material conforming to other internationally accepted standards, which ensure equal or higher quality than the standards mentioned above would be acceptable. In case the Bidders who wish to offer material conforming to the other standards, salient points of difference between the standards adopted and the specific standards shall be clearly brought out in relevant schedule. Four copies of such standards with authentic English Translations shall be furnished along with the offer.

3.0 CLIMATIC CONDITIONS:

3.1 The material called in this specification are required to operate satisfactorily under the following climatic conditions.

i.	Location	:	Andhra Pradesh
ii.	Max ambient air temperature(deg.C)	:	55
iii.	Minimum ambient temperature(deg C)	:	7.5
iv.	Average daily ambient air temp(deg.C)	:	35
v.	Maximum relative humidity(%)	:	74
vi.	Average annual rainfall(mm)	:	925
vii.	Maximum wind measure(kg/m2)	:	200
viii.	Max attitude above mean see level(m)	:	1000
ix.	Ceraunic level (days/year)	:	40 to 50
x.	Seismic level (horizontal acceleration)	:	0.10g

4.0. **PRINCIPAL PARAMETERS**:

4.1 <u>FLOAT CUM BOOST CHARGERS:</u> **(Each FCBC should have following specifications):**

4.1.1 <u>AC input:</u>

i. Voltage: 415V ± 20% 3-Ph AC (From 332V AC to 498V AC) for 48V/ 35A, 50A & 100Achargers

ii. Frequency: 50Hz + 5%

4.1.2 <u>OUTPUT VOLTAGE & CURRENT</u>:

iii. Constant voltage - float mode : 48 to 56V

iv. Constant current - float mode : 25% to 100% of the charging current up to58V

Boost mode : 20 to 60% of the battery charging current for boost charging the battery from 56 to 65V

v. Output current rating : 35A for 48V/35A charger
50A for 48V/50A charger
100A for 48V/100A charger

vi. Current limit

i. Battery path : 10A–15A for 48V/35A charger
10A – 20A for 48V/50A charger
10A – 40A for 48V/100A charger

ii. Total current limit : 105% of the current capacity of the charger

4.1.3 OUTPUT

1. Float Voltage
   a. Load path : 48 to 52V
   b. Battery path : 50 to 56V
2. Boost Voltage
   a. Load path :48-52V
   b. Battery path :56-65V

Suitable number of dropper diodes shall be connected in the load path in a string with a 4position selector switch (0volts to 4volts dropping in steps) to select appropriate number of Diodes to maintain the above load voltage of 48 to 52 V under any battery charging voltage conditions. Dropper Diodes bypass contactor should be provided to feed the battery directly to the load in case of AC supply/charger fail conditions. **The dropper diode bypass contactor should act instantaneously after supply failure.** Manual Selector switch for dropper diode selection shall be provided.

3. Ripple content : ≤ 2% RMS without battery connected
4. Voltage Regulation : +/- 1%
5. Efficiency : Not less than 75% at full load
6. Power Factor : Not less than 0.80 lag at full load

7. Insulation : Not less than 5M ohms with 500V DC Meggar between
   a. DC output terminals and AC input terminals
   b. DC output terminals to body
   c. AC input terminals to body

4.1.4 GENERAL:

a) Cabinet : **(48VDC/35A or 50A or 100A) DC supply system comprising of Float cum boost charger with DC distribution in single cabinet of standard Width, Depth & Hight. Free standing & floor mounting type with bottom cable entry, rear side access for AC supply, load & Battery terminations and Front side access to Controls (load MCB's and other contactors/switches) & Monitoring. The system shall be based on thyristor-controlled design. The charger shall suit to connect the 48V SMF batteries or Lead Acid battery system.**

b)Type of cooling :Natural air cooling
c)Temperature range :0 to 50 deg C. of operation
d)Humidity : 0% to 95%
e) Colour : Silver Ash / Grey
f) Protection : IP-42

5.0 TECHNICAL REQUIREMENTS

5.1. FLOAT CUM BOOST CHARGER (Each FCBC should have following specifications):

5.2.1 The battery charging plant consists of

1) Float cum Boost charger with manual change over facility.
2) D.C. Distribution.

5.2.2 Requirements of Battery charging plant:

The Battery charging plant shall ensure 48V DC supply to PLCC/SDH equipment and Telephone switching equipment and should be able to charge the battery floating across the DC output terminals.

5.2.3 FLOAT CUM BOOST CHARGERS:

In addition to the above, the unit should have facility to Boost charge the 48vstation Batteries under manual control in constant current mode of charging at any rate of charge between 20% and 60% of the rated charging current from a terminal voltage of -58V(2.4V/cell) to a terminal voltage of -65v (up to 2.7v/cell)

5.2.4.0 Under all the battery charging voltage conditions, Load voltage shall be maintained between 48V to 52 V through suitable number of dropper diodes.

5.2.4.1. Output: Constant Voltage mode: Constant voltage in the range of -50 to 56 V DC at any value of load current, from 0 % to 100% of the rated full load. Facility to manually adjust the output voltage (by potentiometer) is to be provided.

5.2.4.2. In case of using with Maintenance free sealed Lead Acid Batteries, constant voltage for charging the Battery will be set in the range 53 to 55V. Battery charging current limit in the Battery path should be provided for this application and this will be set in the range mentioned above. The Unit shall have soft start features and should not have any tendency to hunt irrespective of the input and load conditions.

5.2.4.3. Constant current Mode: In addition to the above, the Unit should have the facility to charge the 48V Stationary Batteries under manual control to a terminal voltage of 2.4V/cell (-58V) from the discharged condition of 1.85V/cell (-44.4V) under constant current mode at any rate of charge between 25% and 100% of the rated charging current.

5.2.4.4.	Current Rating	:	35Amps/50Amps /100 Amps
5.2.4.5.	Output Voltage Drop	:	When the total current(Battery + load) of the charger goes beyond 100% of the rated current, the output terminal voltage should fall sharply at or before 105% of the rated current, to a value below-47v.
5.2.4.6.	Ripple Voltage	:	≤ 2% RMS without battery connected
5.2.4.7	Voltage Regulation	:	+ /- 1% for currents between zero and full load and mains Voltage variations of + /- 10%
5.2.4.8.	i. Efficiency	:	Not less than 75% of full rated load
	ii. Power Factor	:	Not less than 0.8% lag

iii. Total current limit
(Load + battery) : Fixed at 105% of the rated current capacity of the charger.

5.2.4.9. Insulation : Not less than 5 Meg ohms in any of the following cases.

i) between DC output terminals and AC input terminals
ii) between DC output terminals and Earth
iii)between AC input terminals and Earth

5.2.4.10. Protection & Control : The following protection & Controls should be provided

1) AC input Protection
2) HRC Fuses for DC output
3) HRC fuses for SCR's & Diodes
4) DC output adjust potentiometer
5) Battery isolation switch
6) Dropper diodes selector switch
7) Low voltage sensitive relay to isolate load from the battery during main Voltage failure or chargers failure when the battery is drain to the level of -44 .4V(i.e. 1.85 per cell).

5.2.4.11. Equipment over voltage : The FCBC charger should trip and give an EOV indication by LED flash 'ON' and an alarm when the Load Voltage exceeds 55v. It can be switched. "ON" only after manual resetting of the EOV - Reset push button.

5.2.4.12. Supervision and Alarms : The following Visual LED indications should be provided on units with marking.

a) ON-Green
b) Off/mains fail-Amber
c) Fault-Red
d) **Remote Monitoring through Ethernet** : It should be compatible for integration with substation Automation system (SAS) for remote monitoring of charger output, load current, AC input, charger failure and other indications.

5.2.4.13. Fault alarms should be actuated in any one or more of the following conditions.

a) SCR/Diode Fuse failure.
b) DC output High/low beyond limits (EOV/Battery low)
c) Over load
d) A.C. input failure
e) Float cum Boost charger fail
f) Filter capacitor fuse fail

5.2.4.14. (i) Analog DC voltmeter (0-75V Range) to be connected across the battery and load through a switch.
(ii) Analog DC Ammeter 0-100A Range for load on DCDB in addition to charger
(iii) Battery Ammeter (100-0-100A Range FCBC)) (Central zero).
(iv) Input AC supply reading voltmeter 0-500V.

5.2.5 D.C. Distribution

5.2.5.1. Switching arrangement should be essentially provided for

a) Termination of Communication equipments loads
b) Termination of 48V stationery battery/SMF Batteries

5.2.5.2.

a) Adequately rated, insulated copper bus bars should be provided for charging bus and load bus.
b) Switch is to be provided to isolate the battery from the charging bus.
c) Suitable cable glands are to be provided for all the external cables.
d) 24 Nos. of 6A Single Pole MCB's in negative potential line, feeding various DC loads are to be provided. The positive wires are to be brought to a common Bus.
e) **6 Nos. of 230V/AC 6A Three phase outputs with double pole MCB protection in each circuit to be provided.**
f) **Should be provided with High Quality Terminal Block for termination of +ve & - ve leads (2.5 Sq.mm) of feeder cables from Loads**. Also, the internal wiring to the Terminal block from individual load MCB's for negative potential and individual positive potential wiring form common positive bus should be done.

The Guaranteed Technical Particulars to be ensured is enclosed in Annexures. Bidders are requested to indicate Guaranteed Technical Particulars separately in the same format.

ANNEXURE

SCHEDULE OF GUARANTEED AND OTHER TECHNICAL PARTICULRS FOR EACH FLOAT CUM BOOST BATTERY CHARGER

1. Manufacturer's Name, Address & Country :……
2. Supplier's Name & Address : …….
3. Trade Name and model no. : …….
4. Governing Standards : ……
5. Charger dimensions : …….

   (i) Height (mm) : ……
   (ii) Depth (mm) : ……
   (iii) Width (mm) : ……
   (iv) Sheet thickness(mm) : ……..…
6. Charger weight (kg) : ……
7. Charger rated output current : ……
   (i) Float charging mode : ……...
   (ii) Boost charging mode : ……..
8. Load limiter current setting Range (Float mode) : ……...
9. Automatic voltage regulator (Float Mode) : ……..…
   (i) Manufacturer's & Address : ……
   (ii) Type Governing Standard : ……...
   (iii) Percentage stabilization of the output DC Voltage : ……
   (iv) Voltage setting range : …….
   (v) Response time of automatic voltage regulator : …….
10. Manual voltage regulator (Float Mode)
    (i) Manufacturer's name & Address : …….
    (ii) Type
    (iii)Governing Standard : …….
    (iv)Voltage setting range ……....

11. Boost charging current setting range : ……..
12. Boost charging limit setting range : ……..
13. Fail- safe current limits in over load condition
    1. 105% : …….
    2. 125% : ……
    3. 150% : …….
14 Rectifier assembly
    - **Type of semiconductor material : ……..**
    - Rated direct current per cell : ……
    - Rated D.C. Voltage : ……...
    - Rated input Voltage : ……
    - Type of connections of rectifier elements : ……..

15. Rectifier transformer:

Rated KVA & % impedance : …….

One minute power frequency withstand voltage (KV) : …….

16. Instruments:

DC Voltmeter range : ……..

DC Ammeter range : ……..

Dial size : ……..

Accuracy class : …….

17. Connector :

Rated voltage : ……..

Rated current : ……..

No. of power contracts : ……..

No. of type and rating of aux contacts : ……..

Operating coil voltage : ……..

Drop out voltage : ……..

18 Thermal Overload relay :

Tripping current rate : …….

Whether Three phasing protection provided : …….

19 Air Break Switches :

Rated voltage : ……

Rated current : ……

type and material of contacts : …….

20 Fuses:

Rupturing capacity : …….

21. MCB/MCCB

Continuous current capacity(A) : ……..

Rupturing capacity

22. List of documents /Literature enclosed :

23. Whether painting done as per Seven tank method : Yes/No

24 Whether type test certificates enclosed : Yes/No.

Signature of Manufacturer/Supplier
Name & Address

Signature of the bidder:

Place :

Name:

Date :

Designation:

Name of the Company:

ANNEXURE
SCHEDULE OF GUARANTEED AND OTHER TECHNICAL PARTICULRS
DC DISTRIBUTION

I GENERAL

1. Manufacturer's Name & Address : ……..
2. Supplier's Name & Address : ……..
3. Governing standard
4. Capacity Rating
   a) Rated Voltage (Volts) : ……..
   b) Rated current (Amps.) : ……..

5. Symmetrical short circuit withstand current (KA) :…….. and duration

6. Degree of protection as per IS:2147

   a) DC distribution / Switching Cubicle : ……..

   b) Bus-bar chamber. : ……..

7. Cubicle sheet metal thickness for (mm) : …….. front, rear, sides and top.
8. Painting shade as per IS:5
   a) External surface : ……..

   b) Internal surface : ……...

9. Standard height, width & depth (mm) : …….. DC distribution /switching cubicle

10. Width of cable alley (mm) : ……..

11. Earth bus-bar size and material. : ……..

12. Recommended dynamic loading for foundation design. : ……..

13. Approximate weight of DC (kg) distribution : ……..

14. Is extension possible in future. : ……..

15. Lifting hooks provided : Yes/No

16. Type tested & type test certificates enclosed: Yes/No
17. Dimensional outline drawing enclosed: Yes/No

II. BUSBARS AND INSULATORS

1. Material : ……..
2. Bare/Painted/Epoxy insulated sleeved : ……..
3. Minimum clearance (mm) : ……..
4. Continuous current rating at

ambient temperature (50°C.) : ……..

5. Cross-section provided for : ……..
   a) Horizontal busbar (sq.mm) : ……..
   b) Vertical busbar (sq.mm) : ……..

6. Short time current rating for KA 1 sec. : ……..

7. Material of the support insulators. : ……..

8. Temperature rise over ambient temperature(50°C) for continuous current rating (deg.C) : ……..

III. CONTACTORS

1. Voltage rating : ……..
2. No. of poles : ……..
3. Rated voltage of main and auxiliary contacts : ……..
4. Limits of operation
   a) Supply voltage variation (±%) : ……..
   b) Drop out voltage (V) : ……..
5. Rated breaking capacity (kA) : ……..
6. Rated making capacity (kA) : ……..
7. Rated thermal current (kA) : ……..
8. Maximum recommended back up HRC fuse size. : ……..
9. Type tested and type test certificates enclosed : Yes/No

IV. AUXILIARY CONTACTOR

1. Voltage rating (Volts) : ……
2. Coil voltage (Volts) : ……
3. Coil burden (VA) : ……
4. No. of contacts and their rating : ……
5. Type tested and type test certificates enclosed : Yes/No

V. CONTROL/SELECTOR SWITCH

1. Voltage rating : ……
2. Type of handle : ……
3. Number of contacts : ……
4. Rating of contacts : ……
5. No. of positions : ……
6. Type tested and type test certificates enclosed :Yes/No

VI. PUSH BUTTONS

1. Contact type : ……
2. Voltage rating : ……
3. Number of contacts :……..
4. Contact rating : ……..
5. Type tested and type test certificates enclosed : Yes/No

VII. SPACE HEATER

1. Rated voltage (Volts) :……
2. Rated power (kW) :……
3. Thermo state setting range (deg.C) :……
4. Type tested and type test certificates enclosed : Yes/No

VIIII. WIRING AND TERMINAL BLOCKS

A. WIRING

1. Voltage grade : ……..
2. Material of wire and solid/stranded conductor :……
3. Type of insulation :……
4. Minimum size of conductor for
   a) Power wiring (sq.mm) :……
   b) Control wiring (sq.mm) :……

B. TERMINAL BLOCKS

1. Type of control terminal block on :……
   a) Fixed portion and draw out portion :……..
   b) Switchgear and breaker : …….
2. Type of terminal blocks : ……
3. Current rating of terminal blocks :…….
4. Type tested and type test certificates enclosed : Yes/No

IX. INDICATING LAMPS

1. Voltage : ……..
2. Series resistor value and rating : ……..
3. Max. Burning hours : ……..
4. Certificate mark (IS or any other) : ……..

X. INDICATING INSTRUMENTS

Voltmeter (range and accuracy class) : ……..
Ammeter (range and accuracy class) : ……..
Scale details : ……..
Scale cramping, if any : ……..
Type tested and type test certificates enclosed : Yes/No

XI. MCBs/MCCBs

Continuous current rating : ……..
Rated breaking capacity : ……..
Wire size which the terminal can accommodate : ……
Padlocking facility provided : Yes/No
Type tested and type test certificates enclosed : Yes/No

Signature of Manufacturer/Supplier
Name & Address

Signature of the bidder:

Place :

Name:

Date :

Designation:
Name of the Company: